

CSE/ISyE 6740
Computational Data Analysis

Clustering

08/26/2026

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Outline

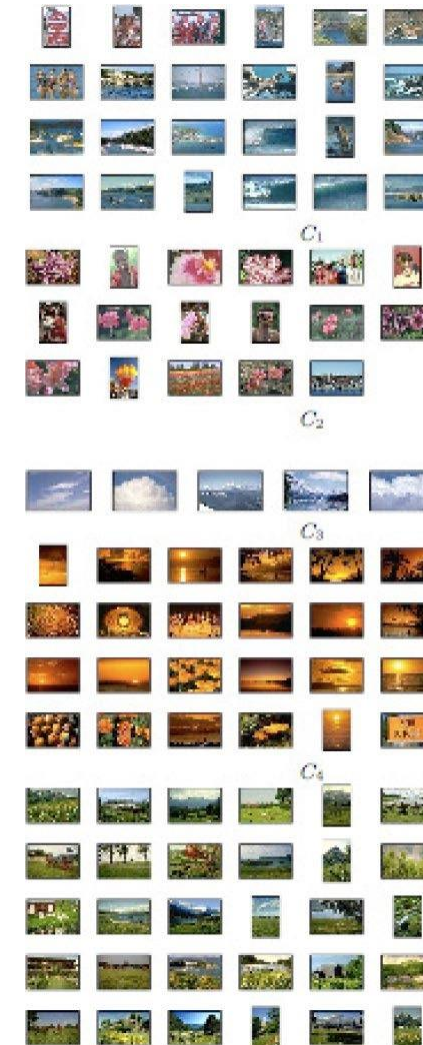
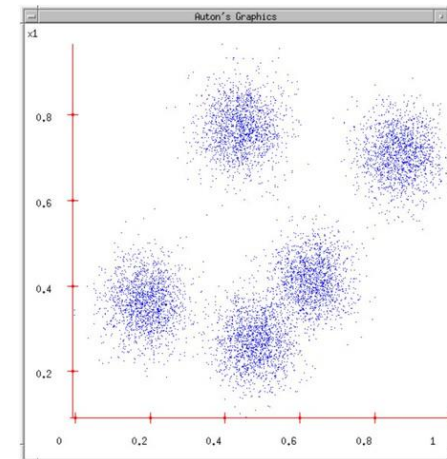
- **Unsupervised learning**
 - **Clustering**
 - Formal statement of clustering problem
 - K-means algorithm
 - Hardness
 - Distance
 - Generalized k-means
 - **Spectral clustering**
 - How to handle network information?
 - Adjacency matrix, graph Laplacian, and eigenvectors as representation!

Clustering

Clustering Images



- **Goal of clustering:** Divide object into groups, and objects within a group are more similar than those outside the group



Cluster Other Things...



Piotr
Pyotr
Petros
Pietro
Pedro
Pierre
Piero
Peter
Peder
Peka
Peadar

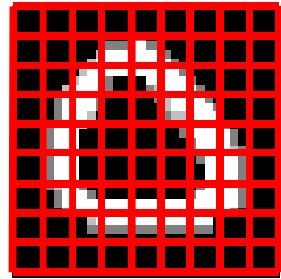


Cluster Handwritten Images

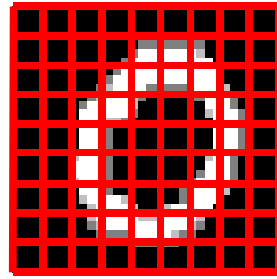
72104149590690159784766540740131
34727121174235124463556041957853
74643070291732977627847361368314
17696054992194873974449254767905
85665781016467317182029955156034
46546545144723271818185084250111
09031642361113952945939036557327
12841733887922415987230442419577
28268577918180301994182129759264
15429204002847124027433003196525
17936420711215339786361381051315
56185179462250656372088541140337
61621928619525442838245031775797
19219292049148184599837600302664
95332391268056663882758961841259
19754089910523789406395213136578
22632654897130383193446421825488
40023277087447969098046063548339
33778087170654380963809968685786
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07944855408210845040613326726931
46259206217341054311749948402451
16471942415538314568941538032512
83440883317359632613607217142221
79611248177480231310770355276692
83522560829288887493066321322930
05781446029147473988471212232303
91740355865267663279117564951334
78911691495406223151203812671623
90122089

How to Represent Objective?

- Represent image as a vector of pixels, and each pixel as a value between 0 and 1.

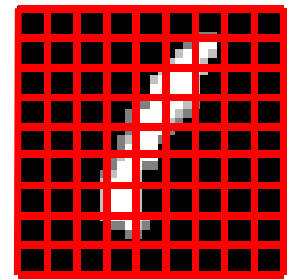


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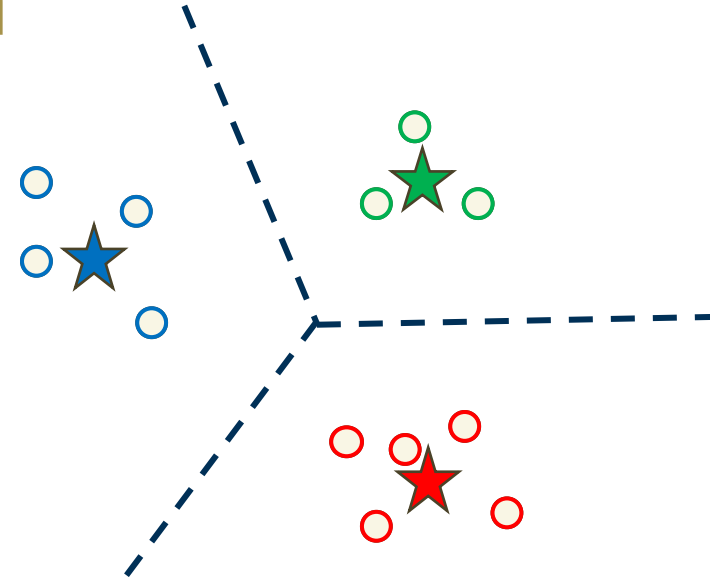
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Formal Statement of Clustering Problem

- Given n data points: $\{x^1, x^2, \dots, x^n\} \in \mathbb{R}^d$
- Find k cluster centers: $\{c^1, c^2, \dots, c^k\} \in \mathbb{R}^d$
- Assign each data point to one cluster: $\pi(i) \in \{1, 2, \dots, k\}$
- Such that the averaged square distances from each data point to its respective cluster center is small, i.e.,

$$\min_{c, \pi} \frac{1}{n} \sum_{i=1}^n \|x^i - c^{\pi(i)}\|^2$$



K-means Algorithm

- Initialize k cluster centers $\{c^1, c^2, \dots, c^k\}$ randomly
- Do
 - Decide the cluster memberships of each data point, x^i , by assigning it to the nearest cluster center (**cluster assignment**)

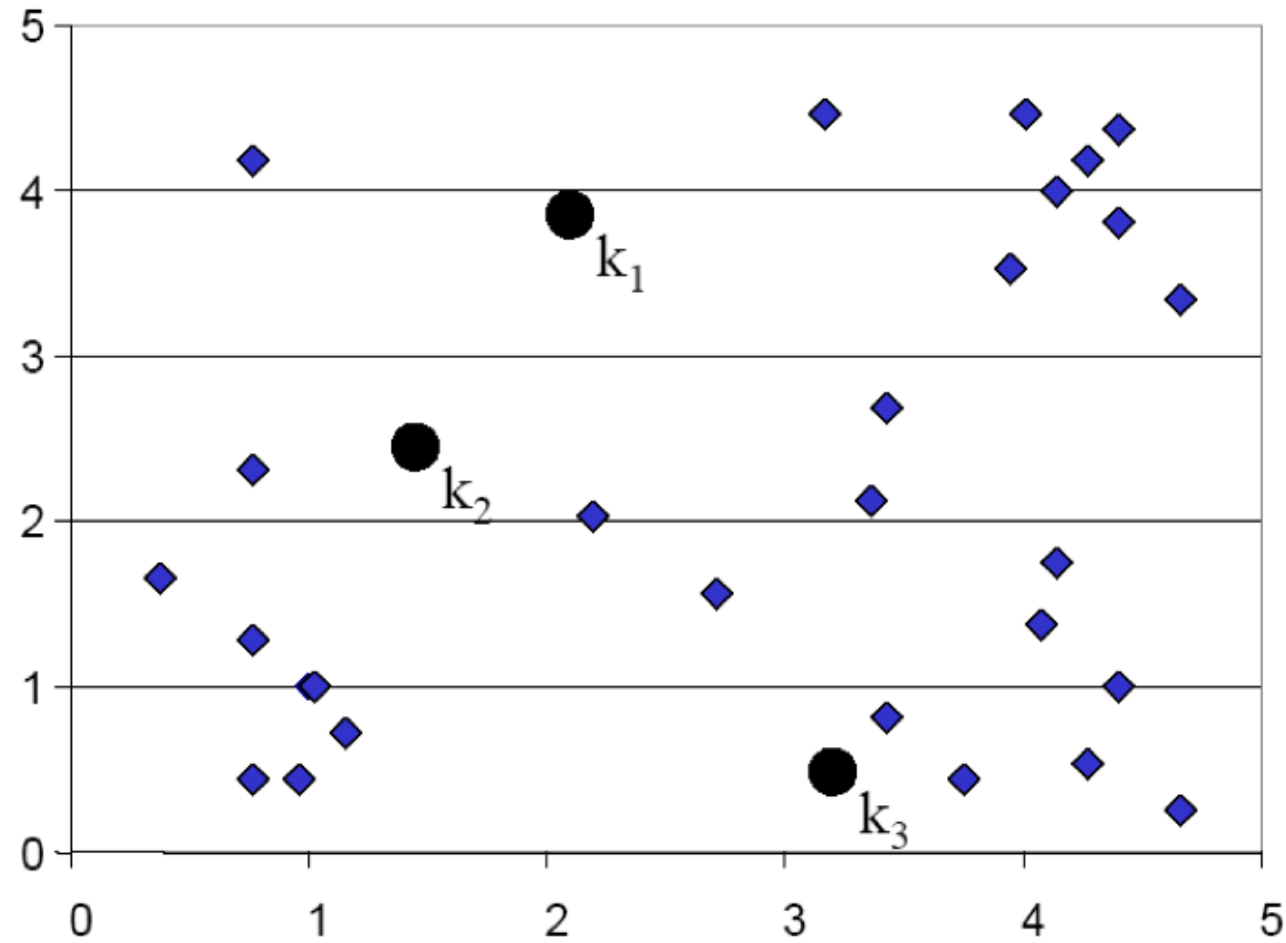
$$\pi(i) = \operatorname{argmin}_{j=1,2,\dots,k} \|x^i - c^j\|^2$$

- Adjust the cluster center (**center adjustment**)

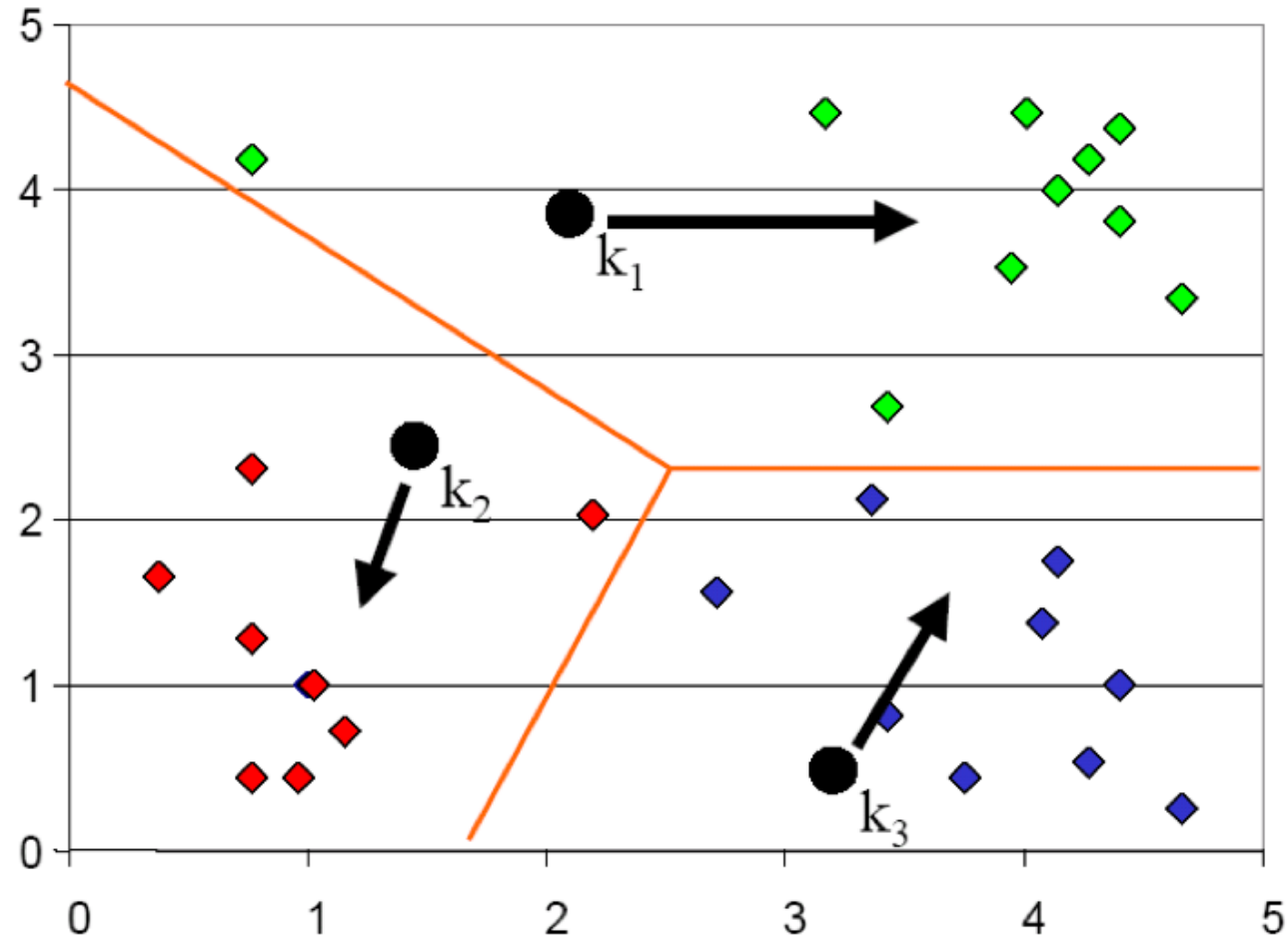
$$c^j = \frac{1}{|\{i: \pi(i) = j\}|} \sum_{i: \pi(i)=j} x^i$$

- While(any cluster center has been changed)

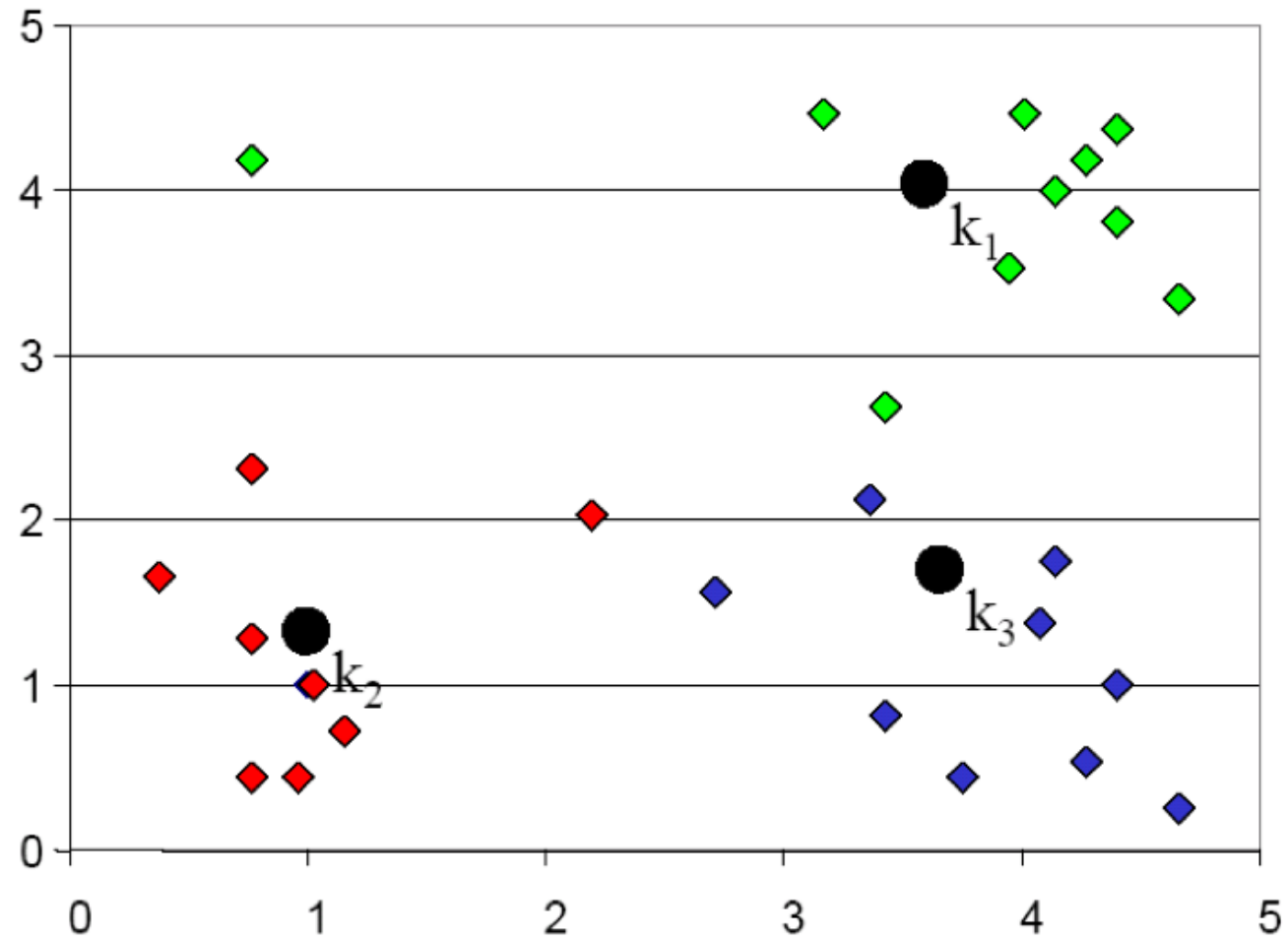
K-means Step 1



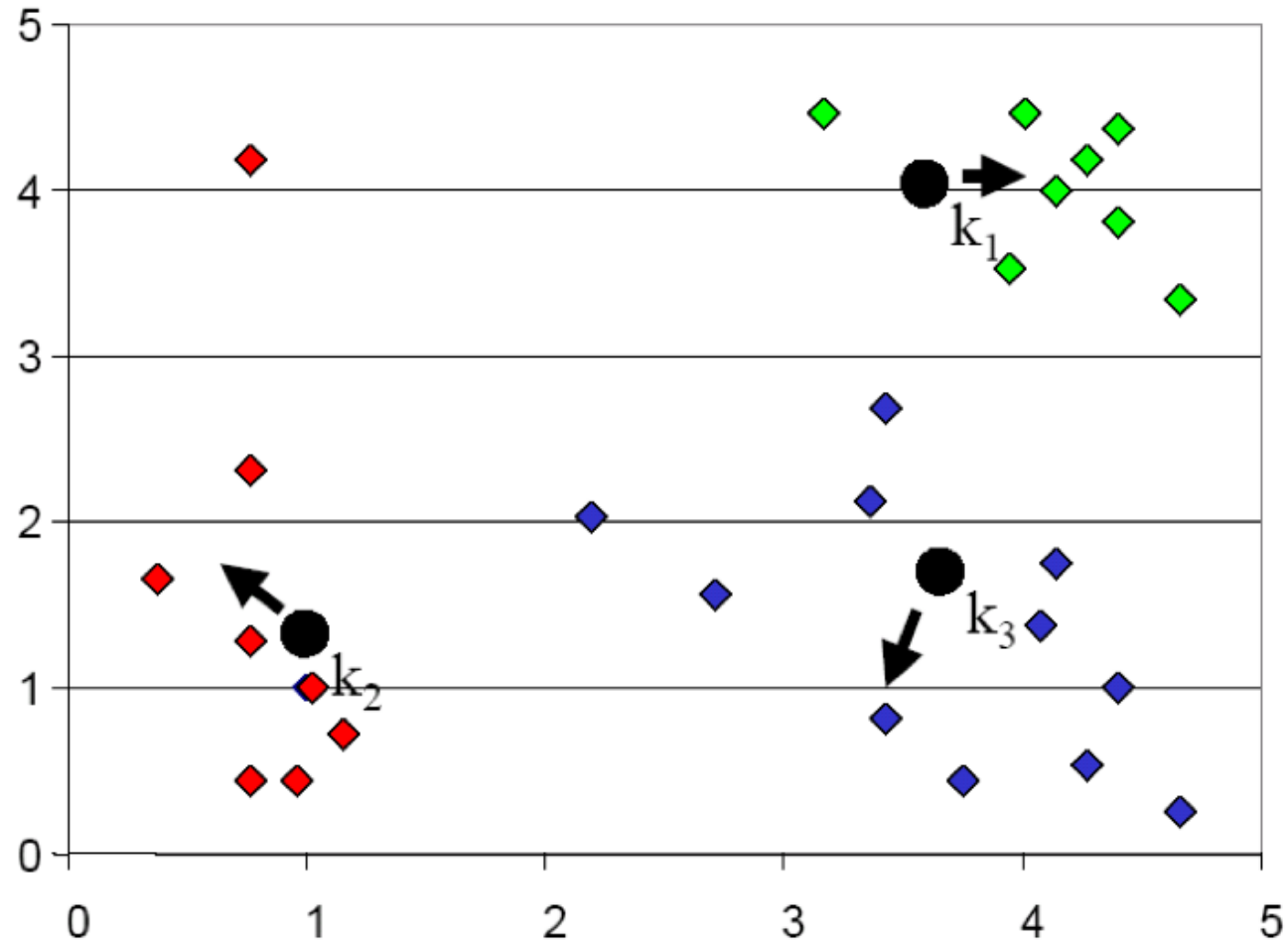
K-means Step 2



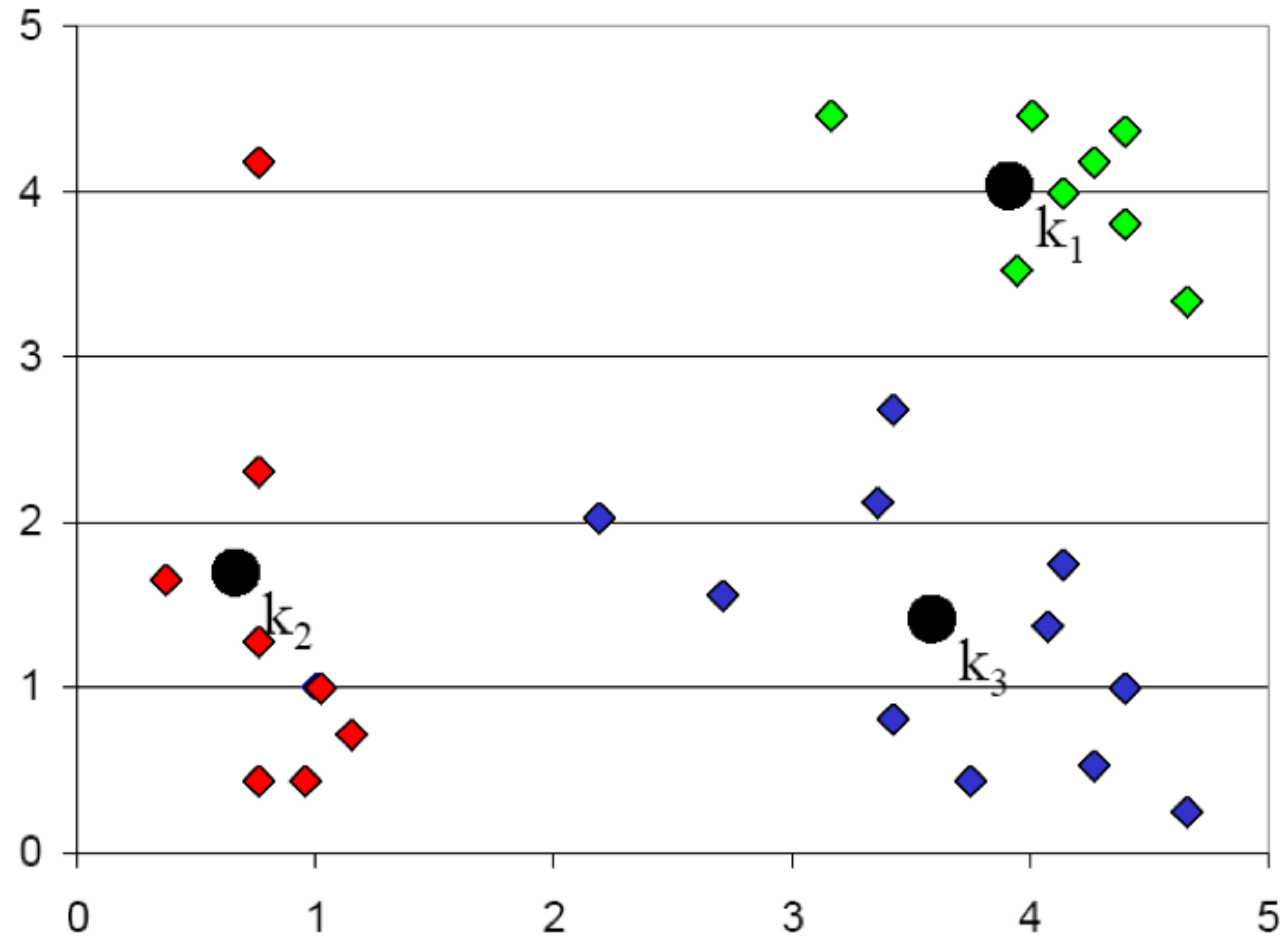
K-means Step 3



K-means Step 4



K-means Step 5



Questions

- Will different initializations lead to different results?
 - Yes
 - No
- Will the algorithm always stop after some iterations?
 - Yes
 - No, we have to set a maximum number of iterations

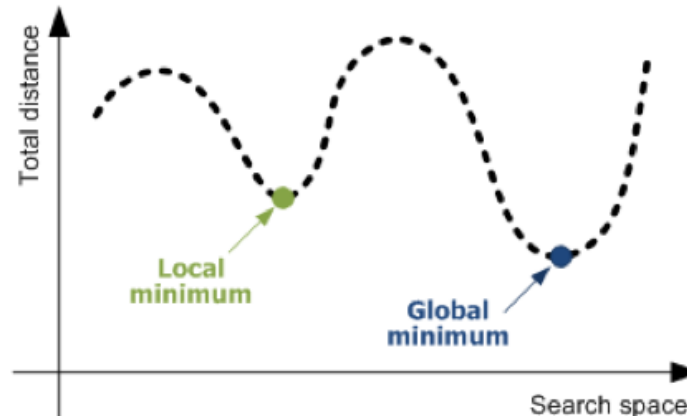
Clustering is NP-hard in General

- Find k cluster centers, $\{c^1, c^2, \dots, c^k\} \in \mathbb{R}^d$, and assign each data point i to one cluster, $\pi(i) \in \{1, 2, \dots, k\}$, to minimize

$$\min_{c, \pi} \frac{1}{n} \sum_{i=1}^n \|x^i - c^{\pi(i)}\|^2$$

NP-hard!

- A search problem over the space of discrete assignments
 - For all n data points together, there are k^n possibility.
 - The cluster assignment determines cluster centers, and vice versa.



Convergence of k-means Clustering

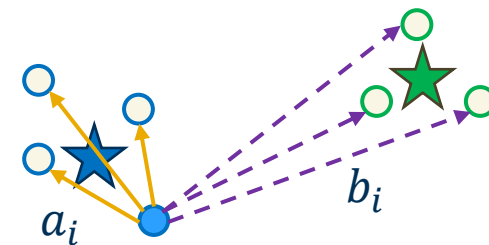
- Will k-means objective oscillate?

$$\frac{1}{n} \sum_{i=1}^n \|x^i - c^{\pi(i)}\|^2$$

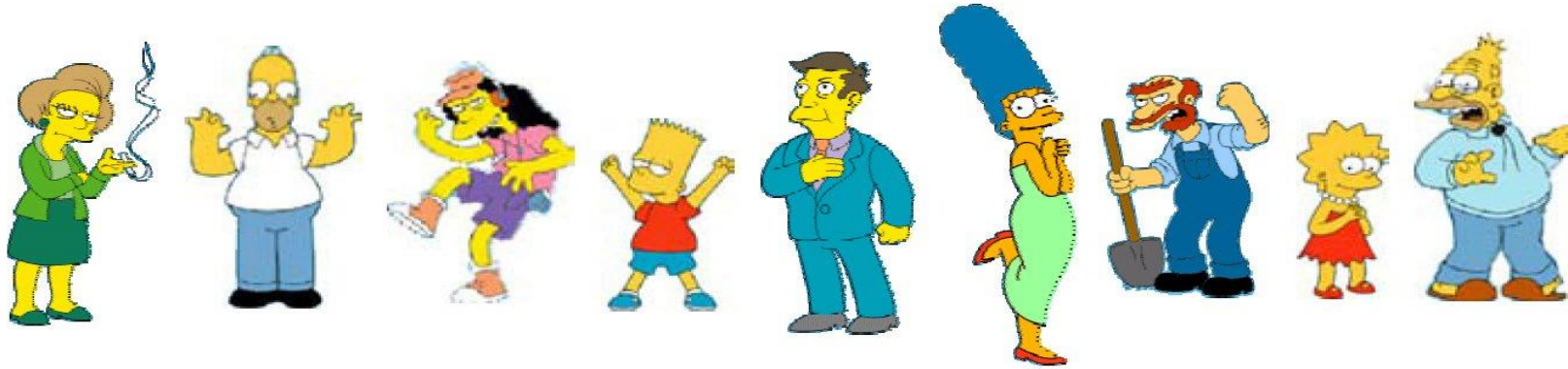
- The minimum value of the objective is finite.
- Each iteration of k-means algorithm decreases the objective.
 - Cluster assignment step decreases the objective
 - $\pi(i) = \operatorname{argmin}_{j=1,2,\dots,k} \|x^i - c^j\|^2$
 - Center assignment step decreases the objective
 - $c^j = \frac{1}{|\{i:\pi(i)=j\}|} \sum_{i:\pi(i)=j} x^i = \operatorname{argmin}_c \sum_{i:\pi(i)=j} \|x^i - c\|^2$

How Many Clusters?

- Fixed a-priori? Data-driven approach?
- Silhouette value: $S_i = \frac{b_i - a_i}{\max(a_i, b_i)} \in [-1, 1]$ (one heuristic)
 - **Distance to closest cluster** b_i : the minimum average distance from the i -th point to points in a different cluster, minimized over clusters.
 - **In-cluster distance** a_i : the average distance from the i -th point to other points in the same cluster as i .
- No gold standard method
 - Often determined by trial-and-error
 - Honestly, I don't use the Silhouette value frequently either

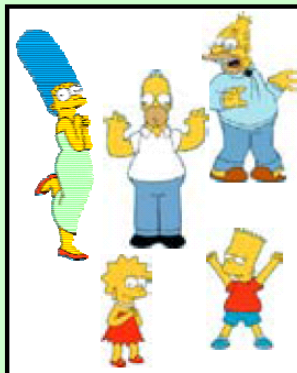


Are These Everything About Clustering?



What is considered similar/dissimilar?

Clustering is subjective



Simpson's Family



School Employees



Females



Males

Object in Real Life

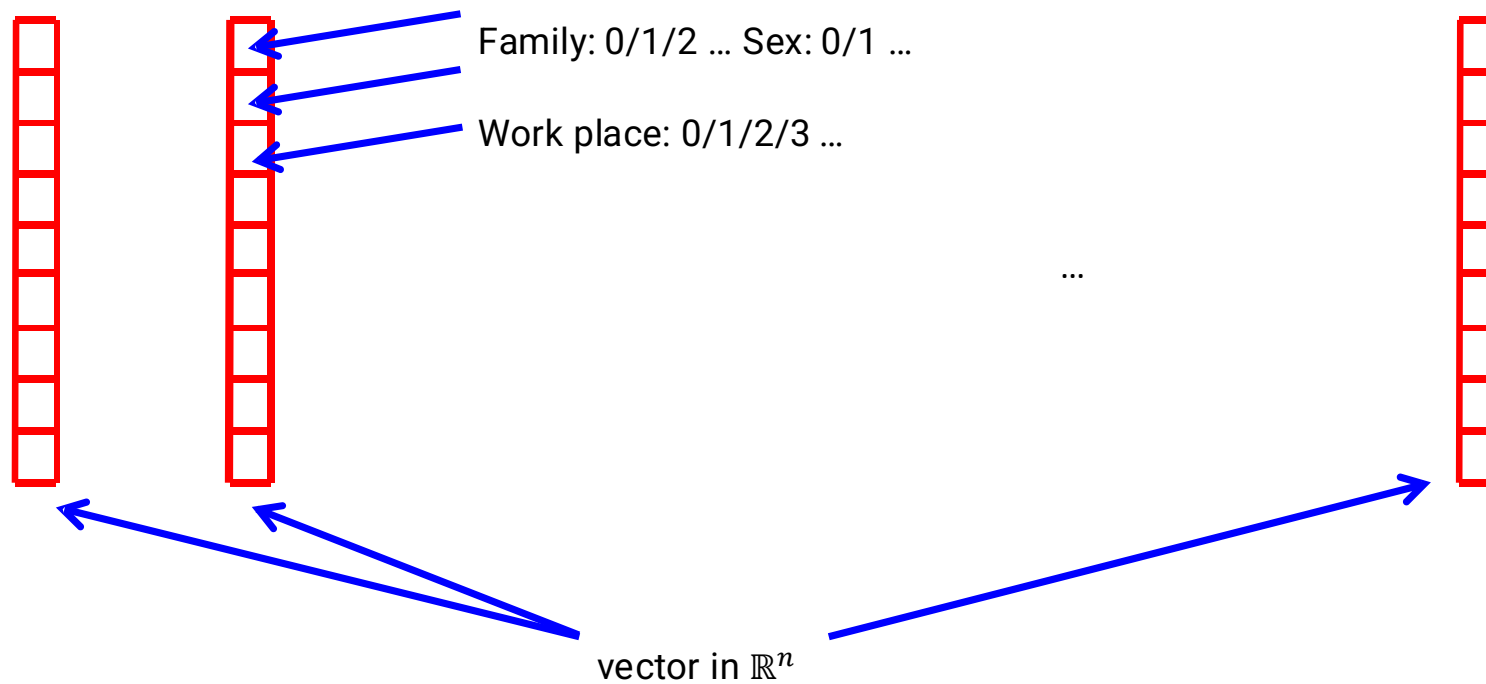
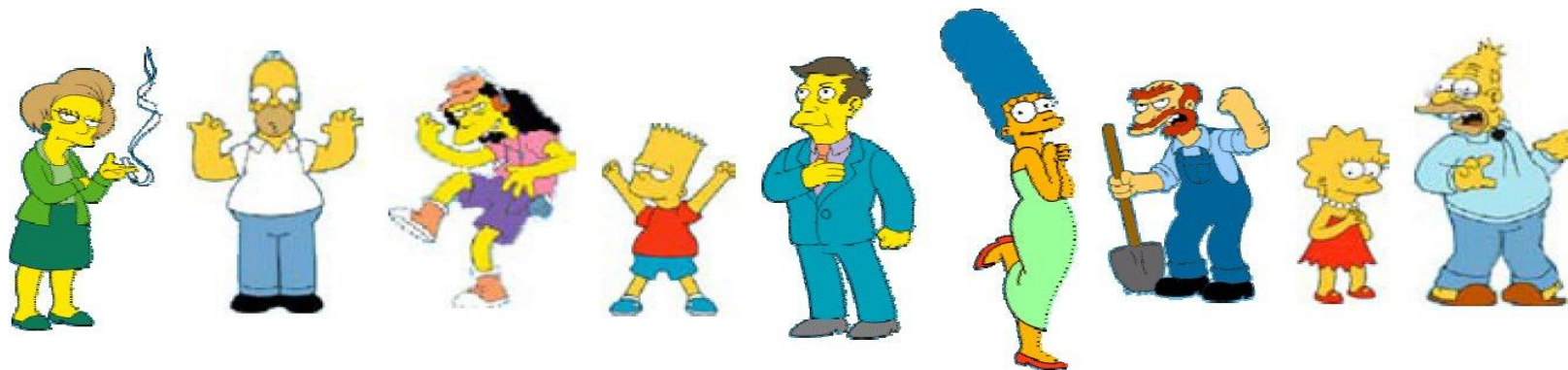
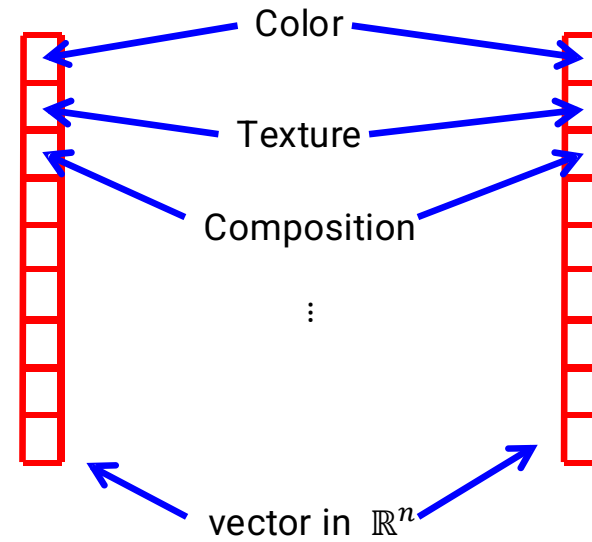


Image of Different Sizes



You Pick Your Similarity/Dissimilarity



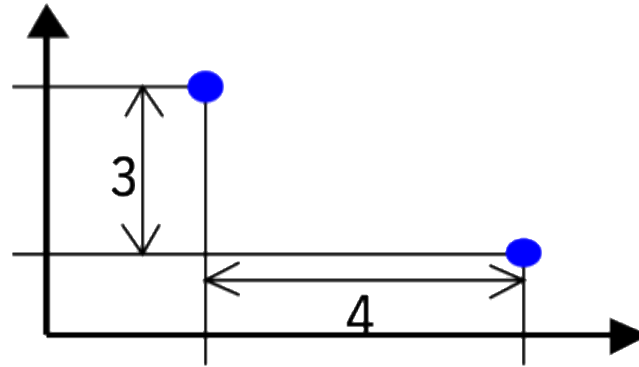
What Similarity/Dissimilarity Function?

- Desired properties of dissimilarity (distance) function $d: X \times X \rightarrow \mathbb{R}^+ \cup \{0\}$
 - **Symmetry:** $d(x, y) = d(y, x)$
 - Otherwise, you could claim “Alex looks like Bob, but Bob doesn’t look like Alex”.
 - **Positive separability:** $d(x, y) = 0$ if and only if $x = y$.
 - Otherwise, there are objects that are different, but you can’t tell apart.
 - **Triangle inequality:** $d(x, y) \leq d(x, z) + d(z, y)$
 - Otherwise, you may have “Alex is very like Bob, Bob is very like Carl, but Alex is very unlike Carl.”

Distance Function for Vectors

- Suppose two data points, both in $\mathbb{R}^d$
 - $x = (x_1, x_2, \dots, x_d)^\top$
 - $y = (y_1, y_2, \dots, y_d)^\top$
- **Euclidean distance:** $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$
- **Minkovski distance:** $d(x, y) = \sqrt[p]{\sum_{i=1}^n (x_i - y_i)^p}$ (or so call p norm)
 - Euclidean distance: $p = 2$
 - Manhattan distance: $p = 1$, $d(x, y) = \sum_i |x_i - y_i|$
 - “inf”-distance: $p = \infty$, $d(x, y) = \max_i |x_i - y_i|$

Distance Example



- Euclidean distance: $\sqrt{3^2 + 4^2} = 5$
- Manhattan distance: $4 + 3 = 7$
- “inf”-distance: $\max\{4, 3\} = 4$

Hamming Distance

- Manhattan distance is also called **Hamming distance** when all the features are binary (or categorical)
- Count the number of differences between two binary (categorical) vectors
- Example, $x, y \in \mathbb{R}^{17}$

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
x	0	1	1	0	0	1	0	0	1	0	0	1	1	1	0	0	1
y	0	1	1	1	0	0	0	0	1	1	1	1	1	1	0	1	1

$$d(x, y) = 5$$

Edit Distance

- Transform one of the objects into the other, and measure how much effort it takes

x = "INTENTION"

y = "EXECUTION"

I	N	T	E	*	N	T	I	O	N
*	E	X	E	C	U	T	I	O	N
d	s	s		i	s				

d: deletion (cost 5)
s: substitution (cost 1)
i: insertion (cost 2)

$$d(x, y) = 5 \times 1 + 1 \times 3 + 2 \times 1 = 10$$

Generalized K-means Clustering

- Initialize k cluster centers $\{c^1, c^2, \dots, c^k\}$ randomly
- Do
 - Decide the cluster memberships of each data point, x^i , by assigning it to the nearest cluster center (**cluster assignment**)
$$\pi(i) = \operatorname{argmin}_{j=1,2,\dots,k} d(x^i, c^j)$$
 - Adjust the cluster center (center adjustment)
$$c^j = \operatorname{argmin}_{v \in \mathbb{R}^d} \sum_{i:\pi(i)=j} d(x^i, v)$$
- While(any cluster center has been changed)

Squared Euclidean distance

$$c^j = \frac{1}{|\{i: \pi(i) = j\}|} \sum_{i:\pi(i)=j} x^i$$

Spectral Clustering

Clustering Nodes in a Network

Visualization tools:

- [Networkx](#)
- [Gephi](#)
- ...

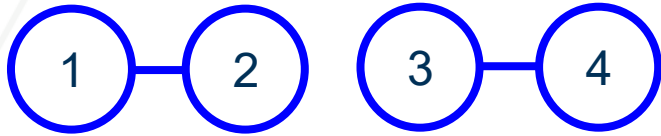


No Feature? Find A Representation!

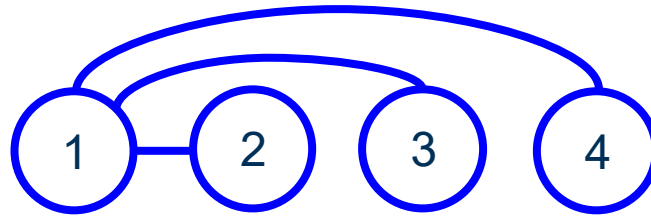
- If we have features associated to each node, you can use them to cluster nodes.
- However, we often don't have features of the nodes in a network. We will need to find a representation to represent the nodes.
- **Adjacency matrix, graph Laplacian**, and their **eigenvectors** are good options for you to handle network data!

Spectral Clustering Algorithm

- **Step 1:** represent the graph as adjacency matrix A , and diagonal matrix D



$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad D = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- **Step 2:** form a special matrix $L = D - A$ (the graph Laplacian)

$$L = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

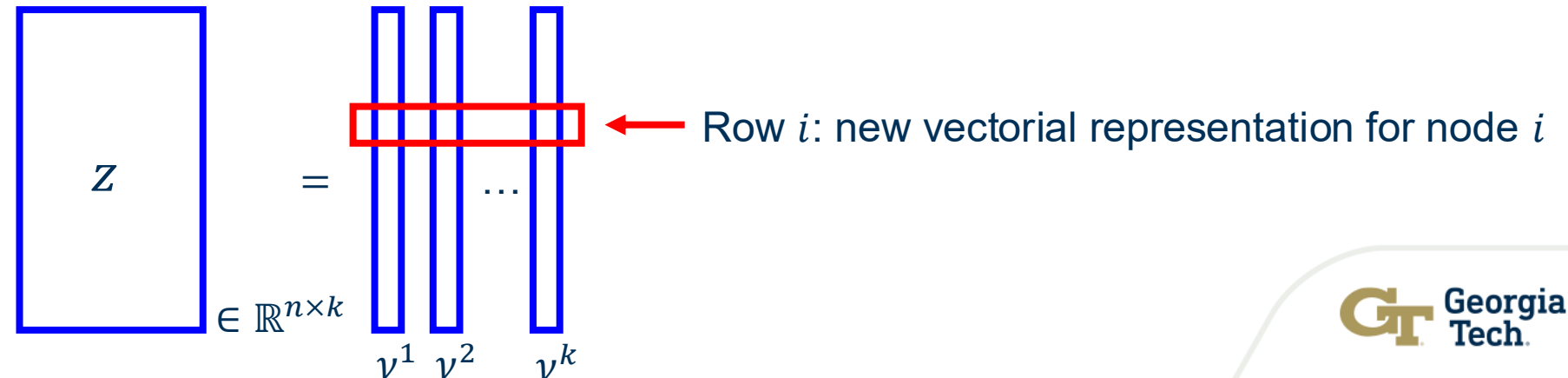
$$L = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

Spectral Clustering Algorithm

- **Step 3:** compute k eigenvectors, $v^1, v^2, \dots, v^k$, of the Laplacian L , corresponding to the k -**smallest** eigenvalues ($k \ll n$).

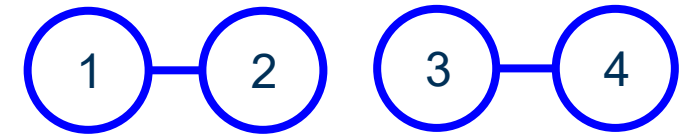
$$Lv^i = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} v^i = \lambda_i v^i$$

- **Step 4:** run k-means algorithm on $Z = (v^1, v^2, \dots, v^k)$ by treating each row as a new data point:



Why This Works?

- **Adjacency matrix** and **Graph Laplacian** include information about neighborhoods in the network (graph).
- Eigenvectors are a good way to represent the matrix and extract the most relevant information, i.e., **Principal Component Analysis**.
 - Eigenvectors tell you information about how nodes are clustered.
 - For connected graphs, eigenvectors with smaller eigenvalues can be used as a compact representation.



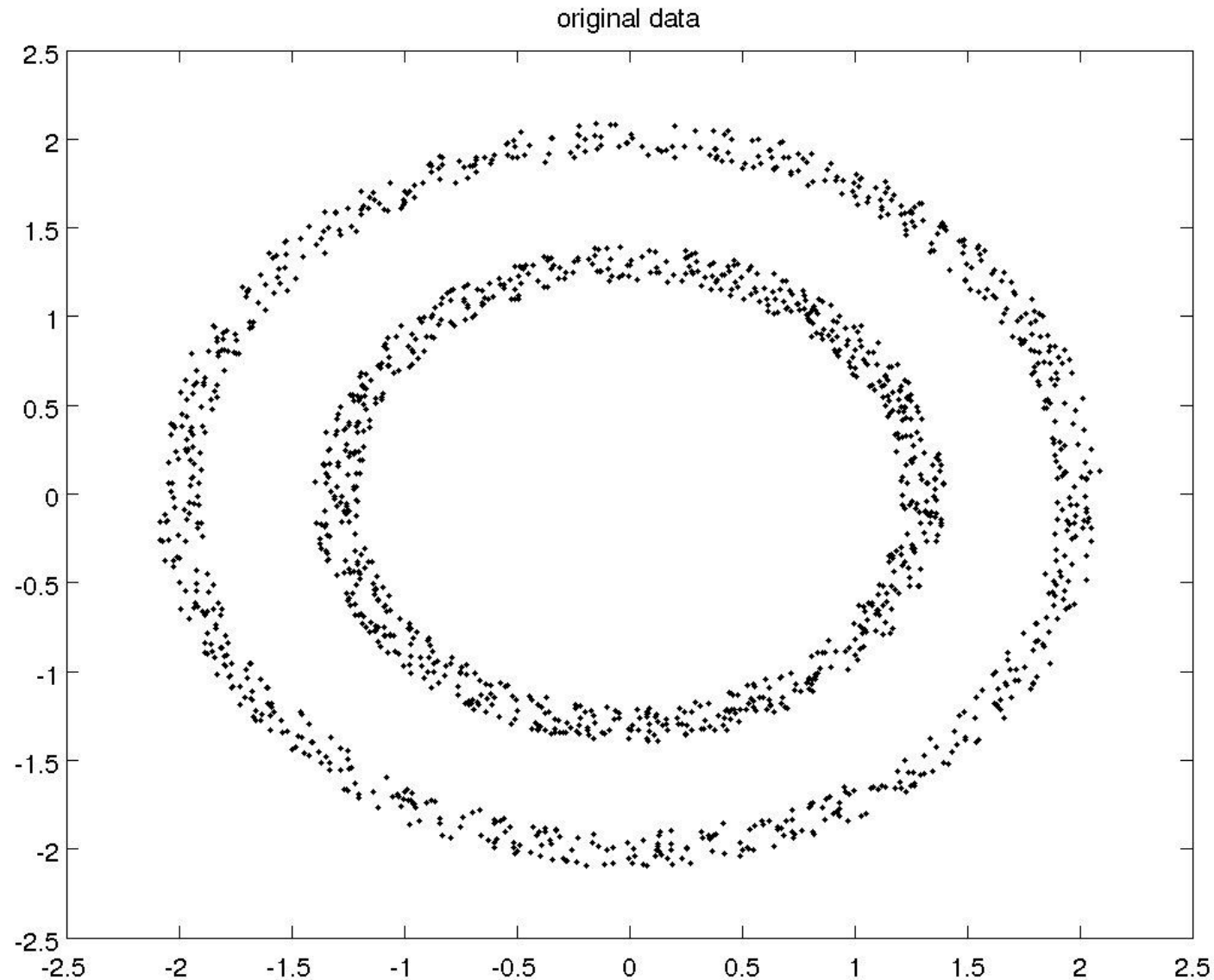
$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$L = D - A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

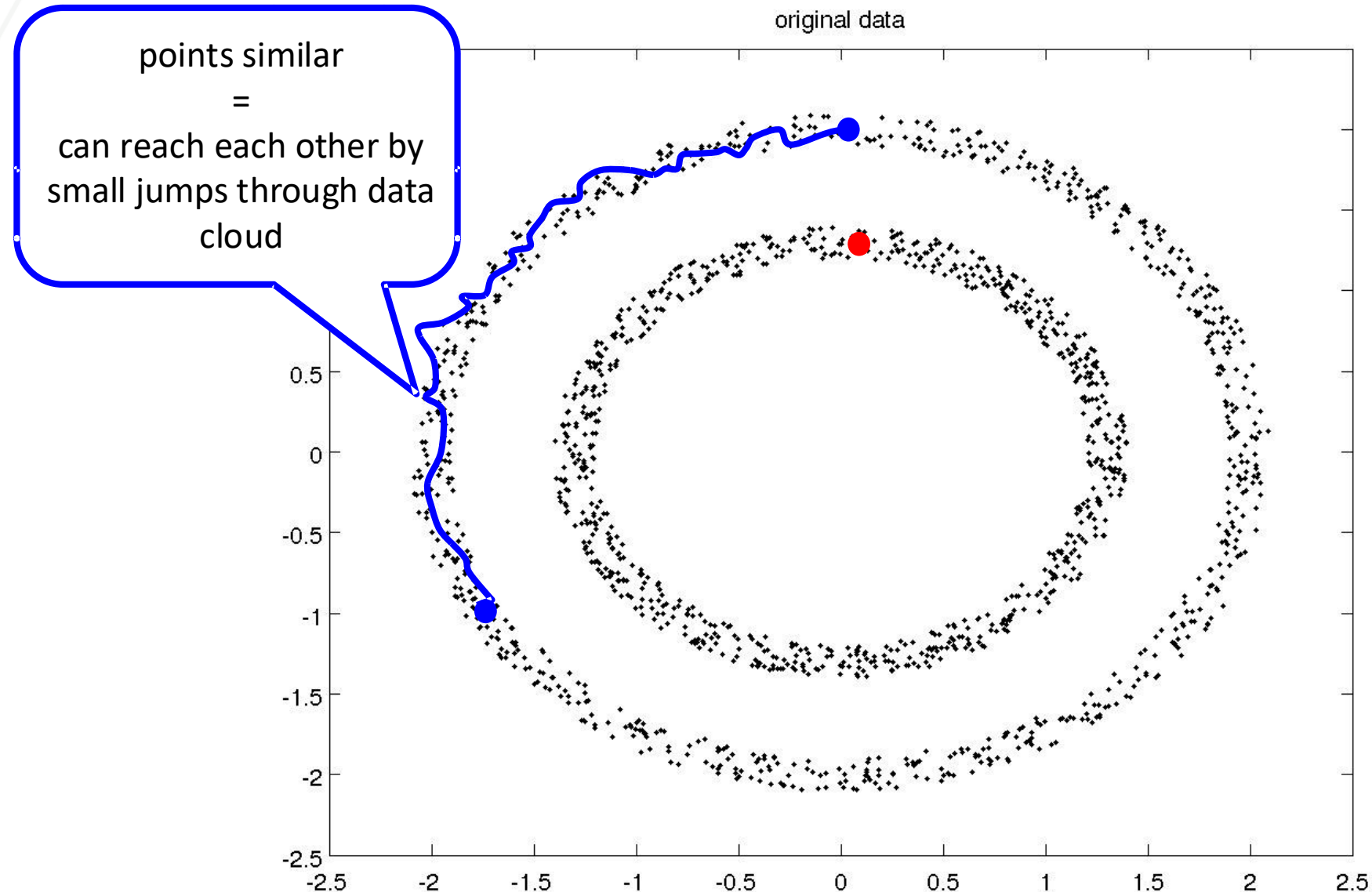
$$Lv^1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$Lv^2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

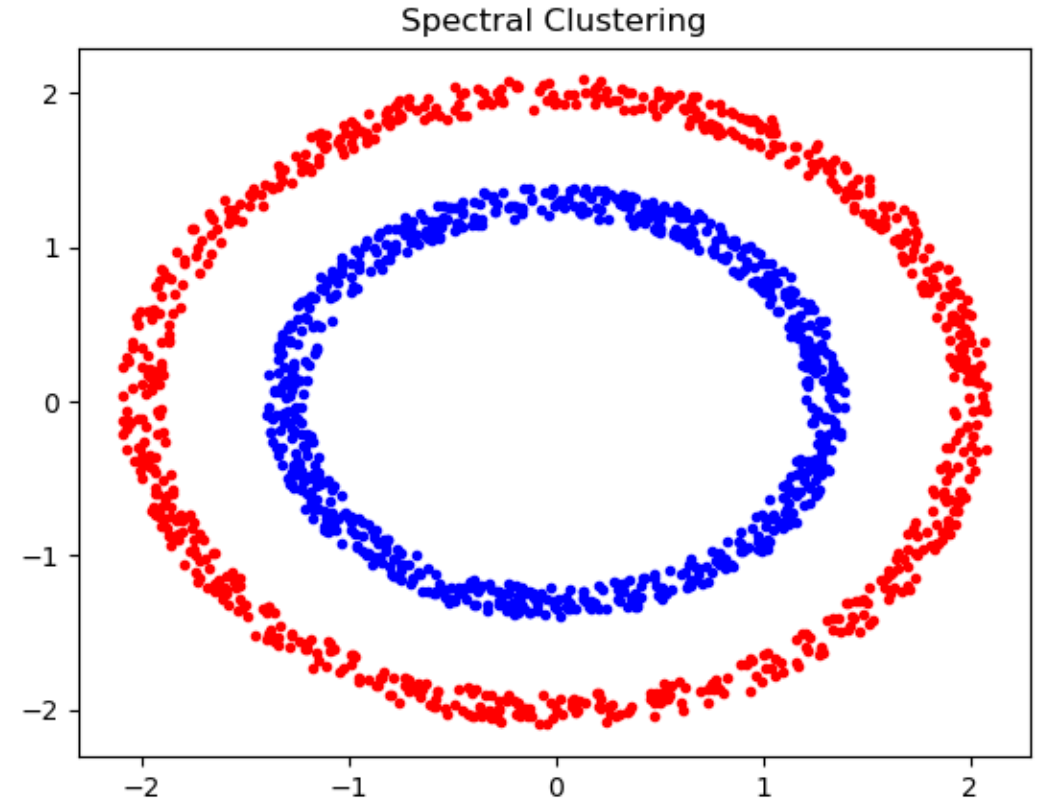
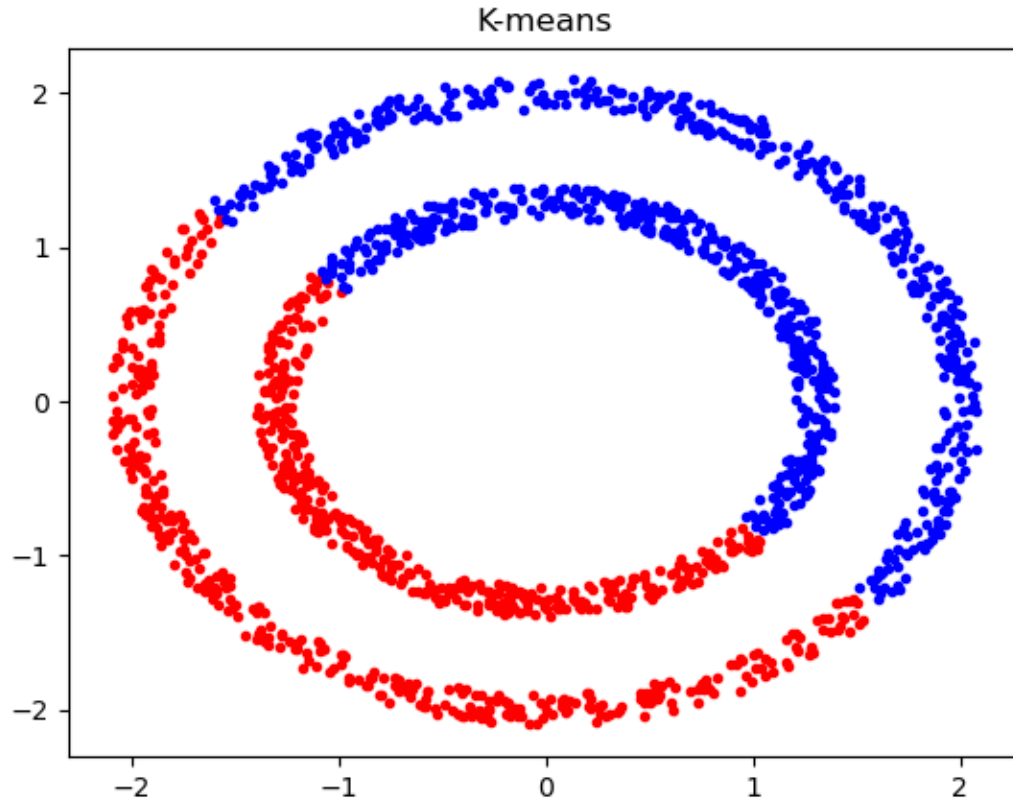
How About this Dataset?



What's a Reasonable Similarity Measure?



Comparison: K-means v.s. Spectral Clustering



Registration

- **Friday** is the registration deadline.
- Work on **Homework 0** earlier to check if you feel comfortable with the prerequisites, and get yourself familiar with Gradescope and autograder.
- If you decide to drop the course, please do so asap so that other people on the waitlist have time to register!
- See you next week!